

## 0.1 Minimal Polynomial

**Definition 0.1.1.** Let  $V$  be a Vector Space over  $F$ , and let  $T : V \rightarrow V$  be a linear operator. A map

$$\varphi_T : F[x] \rightarrow \mathcal{L}(V) : p(x) \mapsto p(T)$$

is called *evaluation* of  $T$ . It is a ring-homomorphism, and  $\ker \varphi_T$  is principal ideal. Moreover,

$$F[x] / \ker \varphi_T \cong F[T]$$

by the first-isomorphism theorem.

Note:  $\mathcal{L}(V) = \{T : V \rightarrow V \mid T \text{ is linear operator}\}$  has algebraic structures:

$(\mathcal{L}(V), +, \cdot)$  is Vector Space over  $F$  and  $(\mathcal{L}(V), +, \circ)$  is a Ring with identity.

Generally,  $(\mathcal{L}(V), +, \circ)$  is non-commutative, and Not domain. For instance, let  $V = F^n$ . Then, for each  $1 \leq j \leq n$ ,

$$A_j : F^n \rightarrow F^n : (a_i)_{i=1}^n \mapsto (0, \dots, a_j, 0, \dots)$$

are not zero maps, but  $E_1 \circ E_2 = \mathbf{0}$ . And ask a toddler on the street, examples for Non-Commutative.

But, the subring

$$F[T] = \{p(T) \mid p(x) \in F[x]\} \subset \mathcal{L}(V)$$

is Commutative Ring with identity, still it is not a domain, in general. Moreover, as a Subspace,

$$F[T] = \text{span}_F \{T^n \mid n = 0, 1, \dots\}$$

If  $V$  is finite-dimensional space, then  $\mathcal{L}(V)$  has a  $n^2$  dimension, therefore

$$T^0, T^1, \dots, T^{n^2}$$

is linearly dependent, this implies there exists a polynomial  $p(x) \in F[x]$  such that  $p(T) = 0$ . In other words,

If  $V$  has a finite dimension, then for linear operator  $T : V \rightarrow V$ ,  $\ker \varphi_T$  is non-zero principal ideal.

In this case, the unique monic polynomial of  $\ker \varphi_T$  is called the *minimal polynomial* for  $T$ .

*Thm.* The characteristic and minimal polynomials for  $T$  have the same roots.

*Proof.* Let  $p$  be the minimal polynomial for  $T$ .

Suppose that  $c \in F$  is root of  $p$ . That is,  $p(c) = 0$ . Then,

$$p = (x - c) \cdot q \quad (\deg q < \deg p)$$

for some  $q \in F[x]$ . Since  $p$  is the minimal,  $q(T) \neq 0$ . Take  $\beta \in V$  s.t.  $q(T)\beta \neq 0$ . Then,  $0 = p(T)(\beta) = (T - cI) \cdot q(T)(\beta)$ . Thus  $T \cdot q(T)(\beta) = c\beta$ ,  $c$  is eigenvalue.

Conversely, let  $c \in F$  be an eigenvalue of  $T$ . That is,  $T\beta = c\beta$  for some nonzero  $\beta \in V$ . Now,  $0 = p(T)(\beta) = p(c)\beta$  and since  $\beta$  is nonzero,  $p(c) = 0$ . ~~QED~~

Conductor.

Def. Let  $W$  be an invariant subspace for  $T: V \rightarrow V$ , and  $\alpha \in V$  be a vector. Define  $T$ -Conductor of  $\alpha$  into  $W$  as;

$$S_T(\alpha; W) \stackrel{\text{def}}{=} \{g \in F[x] \mid g(T)(\alpha) \in W\}.$$

In particular, if  $W = \{0\}$ , the Conductor is called the  $T$ -annihilator of  $\alpha$ .

Lemma. The Conductor  $S(\alpha; W)$  is an Ideal in  $F[x]$ .

Proof. First, if  $T[W] \subseteq W$ , then  $g(T)[W] \subseteq W$ , because for any  $w \in W$ ,  
 $\beta \in W \Rightarrow T^n(\beta) = T^{n-1}(T(\beta)) = \dots = T(T^{n-1}(\beta)) \in W$ .

And since  $W \subseteq V$ , it is closed under the addition and scalar multiplication,  
 $g(T)[W] \subseteq W$ . Now, to show  $S(\alpha; W)$  is an Ideal, first show that it is space.

For any  $f, g \in S(\alpha; W)$  and  $c \in F$ ,  $(cf + g)(T) = cf(T) + g(T)$ , therefore

$$(cf + g)(T)(\alpha) = cf(T)(\alpha) + g(T)(\alpha) \in W$$

Thus it is subspace. Moreover, for any  $f \in F[x]$  and any  $g \in S(\alpha; W)$ ,

$$(fg)(T)(\alpha) = f(T) \underbrace{g(T)(\alpha)}_{\in W} \in W.$$

Thus  $fg \in S(\alpha; W)$ .  $\square$

Note: Given the linear operator  $T$  and  $T$ -invariant subspace  $W$ , and any vector  $\alpha \in V$ ,

$$Q: F[x] \rightarrow V/W: f \mapsto f(T)(\alpha) + W$$

is  $F[x]$ -module Homomorphism, so  $S_T(\alpha; W) = \text{Ker } Q$  is Principal Ideal in  $F[x]$ .

**Lemma.** Let  $V$  be a fin. dim v.s over  $F$ , and let  $T: V \rightarrow V$  be a linear operator. Suppose that the minimal polynomial of  $T$  splits over  $F$  as:

$$p = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k} \quad (c_i \in F)$$

Then, for any proper  $T$ -invariant subspace  $W < V$ , there exists a vector  $\alpha \in V$  s.t.  $\alpha \notin W$  and for some  $c_i$ ,  $(T - c_i I)(\alpha) \in W$ .

**Proof.** Let  $\beta \in V \setminus W$  be given, and let  $q \in F[x]$  be a  $T$ -annihilator of  $\beta$  into  $W$ . Then,  $q$  divides  $p$ . And, since  $\beta \notin W$ ,  $q$  is not constant. Therefore,

$$q = (x - c_1)^{e_1} \cdots (x - c_k)^{e_k} \quad (0 \leq e_i \leq r_i)$$

and at least one of  $e_i$  is positive. Let  $i$  s.t.  $e_i > 0$ . Then,

$$q = (x - c_i) \cdot h \quad \text{where } h = q / (x - c_i) \in F[x].$$

Since  $q$  is  $T$ -annihilator of  $\beta$ ,  $h(T)(\beta) \notin W$ , but  $(T - c_i I)(h(T)(\beta)) = q(T)(\beta) \in W$ .

**Thm.** Let  $V$  be a finite dimensional vector space over  $F$ , and  $T: V \rightarrow V$  be a linear.  $T$  is Triangularizable if and only if minimal polynomial of  $T$  splits over  $F$ .

**Proof.** If  $T$  is triangularizable with respects to a basis  $\beta$ , then the characteristic polynomial is

$$f(x) = \det([T]_{\beta} - xI) = \prod_{i=1}^n (A_{ii} - x) \quad \text{where } A_{ii} \text{ is } i\text{'th diagonal element of } [T]_{\beta}$$

thus, proved. Conversely, choose  $\alpha_1 \in V$  s.t.  $\alpha_1 \notin \{0\}$ ,  $(T - c_1 I)(\alpha_1) \in \{0\}$  for some eigenvalue  $c_1 \in F$  of  $T$ . Choose  $\alpha_2 \in V$  s.t.  $\alpha_2 \notin \langle \alpha_1 \rangle$ ,  $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$ , it is possible because  $\langle \alpha_1 \rangle$  is  $T$ -invariant. Again, choose  $\alpha_3 \in V$  s.t.  $\alpha_3 \notin \langle \alpha_1, \alpha_2 \rangle$ ,  $(T - c_3 I)(\alpha_3) \in \langle \alpha_1, \alpha_2 \rangle$ . It is possible again,  $T(\alpha_2) \in \langle \alpha_1, \alpha_2 \rangle$ .

Now, we obtain  $\{\alpha_1, \dots, \alpha_n\}$  s.t.

$$T(\alpha_1) = c_1 \alpha_1$$

$$T(\alpha_2) = a_{21} \alpha_1 + c_2 \alpha_2$$

⋮

$$T(\alpha_n) = a_{n1} \alpha_1 + a_{n2} \alpha_2 + \dots + c_n \alpha_n$$

Thm. Let  $V$  be a fin. dim. v.s over  $F$ , and  $T: V \rightarrow V$  be a linear operator.

$T$  is diagonalizable if and only if minimal polynomial of  $T$  splits and separable.

Proof. Suppose that  $T$  is diagonalizable. Let  $c_1, \dots, c_k$  be distinct eigenvalues.

Then, for any eigenvector  $d \in V$ , there is a  $c_i \in F$  s.t.  $(T - c_i I)(d) = 0$ .

Therefore, for  $p = (x - c_1) \cdots (x - c_k)$ ,

$$p(T) = (T - c_1 I) \cdots (T - c_k I)(d) = 0$$

because  $\{T - c_1 I, \dots, T - c_k I\}$  commute. Since  $T$  is diagonalizable, for any vector of  $V$  can be represented by linear combination of eigenvectors,

$$p(T) = 0$$

Thus proved. Conversely, let  $W$  be a spanned subspace by all of eigenvectors of  $T$ .

Suppose that  $W \neq V$ . By the construction,  $W$  is  $T$ -invariant. Therefore, by the Lemma, there is  $\alpha \in V \setminus W$  and eigenvalue  $c_0 \in F$  s.t.  $\beta = (T - c_0 I)(\alpha) \in W$ .

Since  $\beta \in W$ ,  $\beta = \beta_1 + \dots + \beta_k$  where  $T(\beta_i) = c_i \beta_i$ ,  $1 \leq i \leq k$ . Therefore, for any  $n \in \mathbb{Z}$

$$h(T)(\beta) = \sum_{i=1}^k h(c_i) \cdot \beta_i \in W.$$

Put  $p = (x - c_0)q$ . Also,  $q - q(c_0) = (x - c_0)h$ . Then,

$$q(T)(\alpha) - q(c_0)(\alpha) = h(T)(T - c_0 I)(\alpha) = h(T)(\beta) \in W$$

And,  $0 = p(T)(\alpha) = (T - c_0 I)q(T)(\alpha) = q(T)(T - c_0 I)(\alpha) \in W$ .

Therefore  $q(c_0)(\alpha) \in W$ , and since  $\alpha \notin W$ ,  $q(c_0) = 0$ . Contradicts to  $p = (x - c_0)^2 q$ .

### Summary.

Let  $T: V \rightarrow V$  be a linear operator on the fin. dim.  $V$  over  $F$ .

For the characteristic polynomial  $f(t) \in F[t]$ , and the minimal polynomial  $p(t) \in F[t]$ ,  $p(t)$  divides  $f(t)$  by the Cayley-Hamilton Thm, and for any  $T$ -invariant  $W \subseteq V$  and  $\alpha \in V$ , every  $T$ -annihilator of  $\alpha$  into  $W$  divides  $p(t)$ . Moreover,  $p(t)$  and  $f(t)$  share the roots, so,

$T$  is triangularizable  $\Leftrightarrow p(t)$  splits over  $F \overset{\text{PIF}}{\Leftrightarrow} f(t)$  splits over  $F \Leftrightarrow T$  has a Jordan form.

$p(t)$  splits over  $F$  and separable  $\Leftrightarrow T$  is diagonalizable  
 $\Rightarrow f(t)$  is separable.

Anyway,  $T$  has a Primary decomposition, moreover,  $T$  has a Rational-Normal form.

Note: let  $\chi(t), m(t) \in F[t]$  be characteristic and minimal polynomial, respectively.  $m(t)$  divides  $\chi(t)$ , because  $\chi(T) = 0 \Leftrightarrow \chi(t) \in (p(t)) \subset F[t] \Leftrightarrow p(t) | \chi(t)$ .  
 $m(t)$  and  $\chi(t)$  have same roots,

But these two facts are independent:

$(t-1) | (t-1)(t-2)$  but these do not share the roots

$(t-1)^2$  and  $(t-1)$  have same roots but  $(t-1)^2 \nmid (t-1)$ .

Calculation.

Let  $D: P_n(\mathbb{R}) \rightarrow P_n(\mathbb{R}): P(x) \mapsto D_x P(x)$ .

Find the minimal polynomial of  $D$  and  $xD$ , and find the Jordan form.

Proof. Clearly, for all  $x^m$  ( $0 \leq m \leq n$ ),

$$D^{n+1}(x^m) = 0, \text{ so } D^{n+1} = 0.$$

And,  $\{I, D, \dots, D^n\}$  is linearly independent, because:

Suppose that  $C_0 I + C_1 D + \dots + C_n D^n = 0$ .

Substituting  $\frac{1}{m!} x^m$  for each  $m = 0, 1, \dots, n$ , then

$$0 = (C_0 I + \dots + C_n D^n) \left( \frac{1}{0!} \cdot x^0 \right) = C_0$$

$$0 = (C_0 D + \dots + C_n D^n) \left( \frac{1}{1!} \cdot x \right) = C_1$$

$\vdots$

So,  $C_0 = \dots = C_n = 0$ . Therefore, the minimal polynomial of  $D$  is  $t^{n+1}$ .

Meanwhile,  $x D(1) = 0$ ,

$$x D(x) = x,$$

$$x D(x^2) = 2x^2,$$

$$\uparrow 3x^3$$

$\vdots$

Therefore,  $[xD]_{\beta} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$ , where  $\beta = \{1, \dots, x^n\}$  is the standard basis for  $P_n(\mathbb{R})$ .

So, the characteristic polynomial is  $t \cdot (t-1) \cdot \dots \cdot (t-n)$ , distinct  $n+1$  roots, the minimal polynomial is same. Now, the Jordan-normal form are

$$[D] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \ddots & 0 \end{bmatrix}, \quad [xD] = \begin{bmatrix} 0 & 0 & 0 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix} \text{ (diagonal)}.$$

Jordan and Rational form of  $3 \times 3$  Matrix.

$$A = \begin{pmatrix} 0 & -3 & 7 \\ -1 & -1 & 5 \\ -1 & -2 & 6 \end{pmatrix}$$

$$t^3 - 4t^2 + 4$$

$$\begin{aligned} f(t) = \det(-A + tI) &= \begin{vmatrix} t & 3 & -7 \\ 1 & t+1 & -5 \\ 1 & 2 & t-6 \end{vmatrix} = t \cdot [(t+1)(t-6) + 10] - (3t-18+14) + (-15+7t+7) \\ &= t(t^2-5t+4) - (3t-4) + (7t-8) \\ &= t^3 - 5t^2 + 4t - 3t + 4 + 7t - 8 \\ &= t^3 - 5t^2 + 8t - 4 \\ &= (t-1)(t-2)^2. \end{aligned}$$

Clearly,  $A - I \neq 0$  and  $A - 2I \neq 0$ .

By the Cayley-Hamilton,  $f(A) = 0$ . So, we will check that either  $(A - I)(A - 2I)$  or  $(A - 2I)^2$  is zero.  $\leftarrow$  has same roots, so we necessary.

$$1). (A - I)(A - 2I) = \begin{pmatrix} -1 & -3 & 7 \\ -1 & -2 & 5 \\ -1 & -2 & 5 \end{pmatrix} \cdot \begin{pmatrix} -2 & -3 & 7 \\ -1 & -1 & 5 \\ -1 & -2 & 6 \end{pmatrix} = \begin{pmatrix} 2+3-7 & \dots & \dots \end{pmatrix} \neq 0.$$

So,  $f(t)$  is the characteristic polynomial and the minimal polynomial.

Now,  $f(t)$  is not separable but splits, so  $A$  has the Jordan form

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

And has a rational form  $\begin{bmatrix} 0 & 0 & 4 \\ 1 & 0 & -8 \\ 0 & 1 & 5 \end{bmatrix}$

Rational form Exercise.

$$\text{Let } A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}$$

$$\begin{aligned} \det(tI - A) &= \begin{vmatrix} t-5 & 6 & 6 \\ 1 & t-4 & -2 \\ -3 & 6 & t+4 \end{vmatrix} = (t-5)(t^2-6t+12) - (6t+24-36) - 3(-12-6t+24) \\ &= (t-5)(t^2-4) - (6t-12) + 3(6t-12) \\ &= t^3-4t-5t^2+20+12t-24 \\ &= t^3-5t^2+8t-4 \\ &= (t-1)(t-2)^2. \end{aligned}$$

The possible:  $(t-1)(t-2)^i$  ( $i=1,2$ ).

$$(A-I)(A-2I) = \begin{pmatrix} 4 & -6 & -6 \\ -1 & 3 & 2 \\ 3 & -6 & -5 \end{pmatrix} \cdot \begin{pmatrix} 3 & -6 & -6 \\ -1 & 2 & 2 \\ 3 & -6 & -6 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{So, } A \sim \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \text{ where } A_1 = \begin{bmatrix} 0 & -2 \\ 1 & 3 \end{bmatrix}, A_2 = [2].$$